ARITHMETIC DYNAMICS ON UNITARY CAYLEY AND GCD GRAPHS: EXACT CONSENSUS RATES AND TWISTED STATES FROM RAMANUJAN SUMS

SAJAD A. SHEIKH

Abstract. The unitary Cayley graph X_n has vertex set $\mathbb{Z}_n$, with $a \sim b$ when $a-b$ is a unit; its eigenvalues are Ramanujan sums. We determine in closed form the two Laplacian eigenvalues that govern linear and nonlinear dynamics on X_n and derive two consequences. First, the optimal constant-step consensus iteration on X_n contracts at the exact rate $\rho^*(X_n) = q/\big(2(p-1)(q-1)+q-2\big)$, where $p<q$ are the two smallest prime divisors of n, and at rate $1/(2p-1)$ when $n=p^a$ with $a \geq 2$. Hence $\rho^*(X_n) \leq 3/5$ for every n, with equality precisely when $6 \mid n$, and the mean value of ρ^* over the integers exists and equals $0.2758\ldots$. Second, we classify the twisted states of the Kuramoto model on X_n completely: a twisted state of additive order M is linearly stable when $M=6$, has identically vanishing Jacobian when $M=4$, and is linearly unstable for every other $M>1$. The stable non-synchronous states are therefore the two hexagonal twists $\pm n/6$, which exist if and only if $6 \mid n$ and attract at exactly half the rate of the synchronous state; the quarter twists are shown to be degenerate saddles of the energy. Extensions to gcd graphs are given. Except for three classical theorems from the stability theory of ordinary differential equations, which are used as explicit assumptions, every result of the paper has been formalised in Lean 4 with Mathlib.

1. Introduction

Spectral graph theory turns questions about dynamics on a network into questions about eigenvalues. For the linear consensus iteration the relevant quantities are the algebraic connectivity λ_2 of Fiedler [6] and the largest Laplacian eigenvalue $\lambda_{\max}$: the best constant step size and the resulting contraction factor are explicit functions of the pair $(\lambda_2, \lambda_{\max})$, as shown by Xiao and Boyd [24]; see also [3, 18]. For networks of identical Kuramoto oscillators [13, 5] on a circulant graph, the linear stability of the twisted states is likewise decided by the Fourier symbol of the graph, a fact going back to Wiley, Strogatz and Girvan [23] for nearest-neighbour rings and developed for general Cayley graphs of cyclic groups by Medvedev and Tang [16]. In both settings a general criterion is available, but what it yields for a given family of graphs depends on how much is known about the spectrum of that family.

This paper studies the family for which the spectrum is purely arithmetic. The *unitary Cayley graph* $X_n = \mathrm{Cay}(\mathbb{Z}_n, \mathbb{Z}_n^\times)$ has the residues modulo n as vertices, two of them being adjacent when their difference is a unit [4, 11]. It is $\varphi(n)$-regular, and its adjacency eigenvalues are the Ramanujan sums $c_n(j)$ [19], which by Hölder's formula depend only on the additive order of j in $\mathbb{Z}_n$ [11]. More generally the *gcd graphs* $X_n(D)$, in which $a \sim b$ when $\gcd(a-b, n)$ belongs to a prescribed set D of divisors, are exactly the integral circulant graphs [21], and their eigenvalues are sums of Ramanujan sums [11]. These graphs have been studied for their energy [9], for perfect state transfer and periodicity of quantum walks [20, 2], and within the general theory of Cayley graph spectra surveyed in [14]. Their role as substrates for classical

2020 *Mathematics Subject Classification.* Primary 05C50; Secondary 11L03, 34D06, 93D50, 68V20.

Key words and phrases. Unitary Cayley graph, gcd graph, Ramanujan sum, algebraic connectivity, consensus, Kuramoto model, twisted state, formal verification.

dynamics appears not to have been examined. In particular, the examples analysed in [16] are Cayley graphs on balls, Paley graphs and complete graphs, and neither unitary Cayley graphs nor Ramanujan sums occur there; on the consensus side, we are not aware of a closed form for the optimal rate on X_n in the literature.

The arithmetic structure of X_n makes exact answers possible where one usually has only bounds. We show that the optimal consensus rate on X_n is a rational function of the two smallest prime divisors of n and of nothing else, and that the complete stability diagram of the twisted states on X_n depends only on the additive order of the twist. The same integer, 6, appears in both results: the consensus rate attains its supremum $3/5$ exactly when $6 \mid n$, and X_n carries a stable non-synchronous twisted state exactly when $6 \mid n$. Both facts have a common source, namely that the largest non-trivial value of $\mu(M)/\varphi(M)$ over $M > 1$ is $1/2$, attained only at $M = 6$.

Contributions. The results of the paper are the following.

(i) *Extremal Ramanujan sums* (Lemma 3.1, Theorem 3.2). For every $n \geq 2$ we determine λ_2 and $\lambda_{\max}$ of the Laplacian of X_n, with multiplicities, in terms of the two smallest prime divisors $p < q$ of n.

(ii) *Exact consensus rate* (Theorem 4.3, Corollary 4.5, Propositions 4.7 and 4.9). The optimal constant-step rate is $q/(2(p-1)(q-1)+q-2)$, or $1/(2p-1)$ for proper prime powers, or 0 for primes. It never exceeds $3/5$, so the number of iterations needed for a given accuracy is bounded independently of n, and its mean value over n exists and is approximately 0.2758.

(iii) *Classification of twisted states* (Theorems 5.4, 5.5 and 5.6, Corollary 5.7). For the Kuramoto model on X_n, a twisted state whose twist has additive order M is linearly stable if $M = 6$, is fully degenerate if $M = 4$, and is linearly unstable otherwise. At the hexagonal states ($M = 6$) the Jacobian equals $-\frac{1}{2}$ times the normalised Laplacian and the decay rate is exactly $K/4$; at the quarter twists ($M = 4$) the Jacobian is the zero matrix and the state is not a local minimum of the energy.

(iv) *GCD graphs* (Propositions 6.1, 6.2 and 6.4). We record the general circulant criteria that the proofs rest on and show that the hexagonal state remains stable on every connected gcd graph $X_n(D)$ with $6 \mid n$ all of whose defining divisors are coprime to 6.

(v) *Formalisation* (Section 8). Except for three classical theorems on the stability of equilibria of ordinary differential equations (Lyapunov's indirect method in its stability and instability forms, and the theorem of Absil and Kurdyka on analytic gradient flows), which enter as explicit hypotheses, every theorem, lemma, proposition and corollary of the paper, and every mathematical assertion made in its remarks, is proved in a Lean 4 development [17, 15] accompanying the paper. The development also proves Hölder's formula for Ramanujan sums and the decomposition (2.5), which are not in Mathlib.

Organisation. Section 2 collects notation and the known spectral facts. Section 3 proves the extremal lemma and the formulas for λ_2 and $\lambda_{\max}$. Section 4 treats consensus and Section 5 the Kuramoto model. Section 6 discusses gcd graphs, Section 7 reports numerical checks, Section 8 describes the Lean development, and Section 9 concludes.

2. Preliminaries

Throughout, $n \geq 2$ is an integer, $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, and $\mathbb{Z}_n^\times$ is its group of units. We write φ for Euler's totient function, μ for the Möbius function, and $\omega(n)$ for the number of distinct prime

divisors of n. For $j \in \mathbb{Z}_n$ the *additive order* of j is

$$\operatorname{ord}(j) = \frac{n}{\gcd(j,n)},$$

a divisor of n; for each divisor M of n there are exactly $\varphi(M)$ elements of order M. The function

$$r(M) = \frac{\mu(M)}{\varphi(M)} \qquad (M \geq 1) \tag{2.1}$$

is used throughout. Note that $r(1) = 1$, $r(2) = -1$, and $|r(M)| \leq 1/2$ for $M \geq 3$, because $\varphi(M) \geq 2$ for $M \geq 3$.

2.1. **Circulant graphs and their spectra.** Let $S \subseteq \mathbb{Z}_n \setminus \{0\}$ satisfy $S = -S$. The circulant graph $\operatorname{Cay}(\mathbb{Z}_n, S)$ has vertex set $\mathbb{Z}_n$ and edges $\{a, a+s\}$, $s \in S$. It is $|S|$-regular, and it is connected if and only if S generates $\mathbb{Z}_n$. For $k \in \mathbb{Z}_n$ let $\chi_k(j) = \mathrm{e}^{2\pi \mathrm{i} kj/n}$. The vectors $\chi_0, \ldots, \chi_{n-1}$ form an orthogonal basis of $\mathbb{C}^n$ consisting of eigenvectors of the adjacency matrix A, with

$$A\chi_k = C_S(k)\,\chi_k, \qquad C_S(k) = \sum_{s \in S} \mathrm{e}^{2\pi \mathrm{i} ks/n} = \sum_{s \in S} \cos\frac{2\pi ks}{n}, \tag{2.2}$$

the last equality because $S = -S$; see [7, Ch. 1] or [14]. The Laplacian $L = |S|I - A$ therefore has eigenvalues $|S| - C_S(k)$, $k \in \mathbb{Z}_n$, and $C_S(-k) = C_S(k)$.

2.2. **Unitary Cayley graphs and Ramanujan sums.** The unitary Cayley graph is $X_n = \operatorname{Cay}(\mathbb{Z}_n, \mathbb{Z}_n^\times)$. It is connected and $\varphi(n)$-regular, and by (2.2) its adjacency eigenvalues are the Ramanujan sums

$$c_n(j) = \sum_{\substack{1 \leq s \leq n \\ \gcd(s,n)=1}} \mathrm{e}^{2\pi \mathrm{i} js/n}.$$

Hölder's evaluation [8, §16.6], used for X_n in [11], gives

$$c_n(j) = \mu\big(\operatorname{ord}(j)\big)\,\frac{\varphi(n)}{\varphi\big(\operatorname{ord}(j)\big)} = \varphi(n)\,r\big(\operatorname{ord}(j)\big). \tag{2.3}$$

Consequently the Laplacian $L = \varphi(n)I - A$ of X_n has the eigenvalues

$$\lambda(j) = \varphi(n)\big(1 - r(\operatorname{ord}(j))\big), \qquad j \in \mathbb{Z}_n, \tag{2.4}$$

and the eigenvalue attached to a divisor M of n occurs with multiplicity at least $\varphi(M)$.

2.3. **GCD graphs.** Let D be a set of divisors of n with $n \notin D$. The gcd graph $X_n(D)$ is $\operatorname{Cay}(\mathbb{Z}_n, S_D)$ with $S_D = \{s \in \mathbb{Z}_n : \gcd(s,n) \in D\}$; thus $X_n = X_n(\{1\})$. The elements s with $\gcd(s,n) = d$ are $s = dt$ with t a unit modulo n/d, whence [11]

$$C_{S_D}(k) = \sum_{d \in D} c_{n/d}(k). \tag{2.5}$$

All eigenvalues of $X_n(D)$ are integers, and every integral circulant graph is a gcd graph [21]. The graph $X_n(D)$ is connected if and only if the elements of D have greatest common divisor 1.

3. Extremal Ramanujan sums and the Laplacian spectrum

Before presenting the main theorems, we establish the auxiliary result on which everything else depends. For $n \geq 2$ let $p = p(n)$ denote the smallest prime divisor of n and, when $\omega(n) \geq 2$, let $q = q(n)$ denote the second smallest *distinct* prime divisor.

Lemma 3.1 (Extremal values of r on divisors)**.** *Let $n \geq 2$ and let M range over the divisors of n with $M > 1$.*

(i) $r(M) \geq -\dfrac{1}{p-1}$, *with equality if and only if $M = p$.*

(ii) *If $\omega(n) \geq 2$, then $r(M) \leq \dfrac{1}{(p-1)(q-1)}$, with equality if and only if $M = pq$.*

(iii) *If $n = p^a$ with $a \geq 2$, then $r(M) \leq 0$, with equality if and only if $M = p^k$ for some $2 \leq k \leq a$.*

(iv) *If $n = p$ is prime, then $M = p$ and $r(M) = -\dfrac{1}{p-1}$.*

Proof. If M is not squarefree then $r(M) = 0$. If $M = \ell_1 \ell_2 \cdots \ell_t$ with primes $\ell_1 < \ell_2 < \cdots < \ell_t$, then $r(M) = (-1)^t / \prod_i (\ell_i - 1)$. Every ℓ_i divides n, so $\ell_1 \geq p$, and when $t \geq 2$ we have $\ell_2 > \ell_1 \geq p$, so ℓ_2 is a prime divisor of n exceeding p and thus $\ell_2 \geq q$.

(i) The value $r(M)$ is negative only if M is squarefree with t odd. Then $\prod_i (\ell_i - 1) \geq \ell_1 - 1 \geq p - 1$, so $r(M) \geq -1/(p-1)$. Equality requires $\prod_i (\ell_i - 1) = p - 1$. If $t \geq 3$, then $\ell_2 - 1 \geq 2$ and $\ell_3 - 1 \geq 4$, so $\prod_i (\ell_i - 1) \geq 8(\ell_1 - 1) > p - 1$. Hence $t = 1$ and $\ell_1 = p$, that is, $M = p$. Conversely $r(p) = -1/(p-1)$.

(ii) The value $r(M)$ is positive only if M is squarefree with $t \geq 2$ even. Then $\prod_i (\ell_i - 1) \geq (\ell_1 - 1)(\ell_2 - 1) \geq (p-1)(q-1)$, which gives the bound. Equality forces $\ell_1 = p$, $\ell_2 = q$, and $\prod_{i \geq 3} (\ell_i - 1) = 1$; since $\ell_i \geq 5$ for $i \geq 3$, this means $t = 2$ and $M = pq$. Conversely $r(pq) = 1/((p-1)(q-1)) > 0$, and $pq \mid n$.

(iii) The divisors $M > 1$ of p^a are $p, p^2, \ldots, p^a$, with $r(p) = -1/(p-1) < 0$ and $r(p^k) = 0$ for $k \geq 2$.

(iv) This is immediate. □

We now present the first main result.

Theorem 3.2 (Extreme Laplacian eigenvalues of X_n)**.** *Let $n \geq 2$ and let $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n = \lambda_{\max}$ be the Laplacian eigenvalues of X_n. Then*

$$\lambda_{\max} = \frac{p}{p-1}\,\varphi(n), \tag{3.1}$$

attained exactly at the $p - 1$ indices j of order p, and

$$\lambda_2 = \begin{cases} \varphi(n)\Big(1 - \dfrac{1}{(p-1)(q-1)}\Big), & \omega(n) \geq 2, \\ \varphi(n), & n = p^a,\ a \geq 2, \\ n, & n = p. \end{cases} \tag{3.2}$$

The eigenvalue λ_2 has multiplicity $(p-1)(q-1)$ in the first case, $p^a - p$ in the second, and $n - 1$ in the third.

Proof. By (2.4) the non-zero-index eigenvalues are $\varphi(n)(1 - r(M))$ with $M = \operatorname{ord}(j) > 1$, each divisor $M > 1$ of n contributing $\varphi(M)$ indices. The largest eigenvalue corresponds to the minimum of r, which by Lemma 3.1(i) is $-1/(p-1)$, attained only at $M = p$; this gives (3.1)

and the multiplicity $\varphi(p) = p-1$. The smallest eigenvalue among the indices $j \neq 0$ corresponds to the maximum of r over $M > 1$. In the three cases of (3.2) this maximum is $1/((p-1)(q-1))$, 0 and $-1/(p-1)$ by parts (ii), (iii) and (iv) of Lemma 3.1, attained at $M = pq$, at $M \in \{p^2, \dots, p^a\}$ and at $M = p$ respectively. The multiplicities are $\varphi(pq) = (p-1)(q-1)$, $\sum_{k=2}^{a} \varphi(p^k) = p^a - p$ and $\varphi(p) = p - 1 = n - 1$. In the first case $(p-1)(q-1) \geq 2$, so $\lambda_2 > 0$; in the other cases $\lambda_2 > 0$ trivially. In the third case $\lambda_2 = \varphi(p)\, p/(p-1) = p$. □

Corollary 3.3 (Spectral ratio)**.** *For $\omega(n) \geq 2$,*

$$\frac{\lambda_2}{\lambda_{\max}} = \frac{(p-1)(q-1)-1}{p\,(q-1)},$$

and for $n = p^a$ with $a \geq 2$ the ratio is $(p-1)/p$. In particular $\lambda_2/\lambda_{\max} \geq 1/4$ for every n, with equality if and only if $6 \mid n$.

Proof. The formulas follow by dividing (3.2) by (3.1). For $\omega(n) \geq 2$ write $u = p - 1 \geq 1$ and $v = q - 1 \geq 2$; then $\lambda_2/\lambda_{\max} = (uv-1)/((u+1)v)$, which is increasing in u and in v, so its minimum over the admissible pairs is at $(u, v) = (1, 2)$, where it equals $1/4$. For prime powers the ratio is at least $1/2$, and for primes it is 1. □

4. Consensus on unitary Cayley graphs

Consider the linear iteration

$$x(t+1) = (I - \varepsilon L)\, x(t), \qquad t = 0, 1, 2, \dots, \tag{4.1}$$

on a connected graph with Laplacian L and eigenvalues $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n$, where $\varepsilon > 0$ is a constant step. Write $\bar{x} = \frac{1}{n}\mathbf{1}^{\top} x(0)$. Since L is symmetric with $L\mathbf{1} = 0$, the average is conserved and the error $e(t) = x(t) - \bar{x}\mathbf{1}$ lies in $\mathbf{1}^{\perp}$ and satisfies $\|e(t)\| \leq \rho(\varepsilon)^t \|e(0)\|$, where

$$\rho(\varepsilon) = \max_{2 \leq i \leq n} |1 - \varepsilon\lambda_i| = \max\bigl\{|1 - \varepsilon\lambda_2|,\, |1 - \varepsilon\lambda_n|\bigr\},$$

with equality for suitable initial data. The following lemma is due to Xiao and Boyd [24]; we include the short proof because it is one of the statements formalised in Section 8.

Lemma 4.1 (Optimal constant step)**.** *Let $0 < a \leq b$. For every $\varepsilon \in \mathbb{R}$,*

$$\max\bigl\{|1 - \varepsilon a|,\, |1 - \varepsilon b|\bigr\} \;\geq\; \frac{b-a}{b+a},$$

and for $\varepsilon^* = 2/(a+b)$ one has $|1 - \varepsilon^*\lambda| \leq (b-a)/(b+a)$ for all $\lambda \in [a, b]$. Hence $\min_{\varepsilon} \rho(\varepsilon) = (\lambda_n - \lambda_2)/(\lambda_n + \lambda_2)$, attained at $\varepsilon^* = 2/(\lambda_2 + \lambda_n)$.

Proof. Put $\rho = (b-a)/(b+a)$. If both $|1 - \varepsilon a| < \rho$ and $|1 - \varepsilon b| < \rho$, then $1 - \varepsilon a < \rho$ and $\varepsilon b - 1 < \rho$. Multiplying the first inequality by $b > 0$, the second by $a > 0$ and adding gives $b - a < \rho(a + b) = b - a$, a contradiction. For the second claim, $(a+b)(1 - \varepsilon^*\lambda) = a + b - 2\lambda$ lies between $a - b$ and $b - a$ when $a \leq \lambda \leq b$; dividing by $a + b$ gives $|1 - \varepsilon^*\lambda| \leq \rho$. □

Definition 4.2. The *optimal consensus rate* of a connected graph G is $\rho^*(G) = (\lambda_{\max} - \lambda_2)/(\lambda_{\max} + \lambda_2)$.

TABLE 1. Spectral data, optimal step, optimal consensus rate and stable non-synchronous twists for some unitary Cayley graphs (Theorems 3.2, 4.3 and Corollary 5.7).

n	(p,q)	$\varphi(n)$	λ_2	$\lambda_{\max}$	ε^*	$\rho^*(X_n)$	stable twists $m \neq 0$
6	$(2,3)$	2	1	4	2/5	3/5	1, 5
7	—	6	7	7	1/7	0	none
10	$(2,5)$	4	3	8	2/11	5/11	none
12	$(2,3)$	4	2	8	1/5	3/5	2, 10
15	$(3,5)$	8	7	12	2/19	5/19	none
16	—	8	8	16	1/12	1/3	none
30	$(2,3)$	8	4	16	1/10	3/5	5, 25
35	$(5,7)$	24	23	30	2/53	7/53	none
49	—	42	42	49	2/91	1/13	none

Theorem 4.3 (Exact consensus rate on X_n)**.** *Let $n \geq 2$, and let $p < q$ be the two smallest distinct prime divisors of n when $\omega(n) \geq 2$. Then*

$$\rho^*(X_n) = \begin{cases} \dfrac{q}{2(p-1)(q-1)+q-2}, & \omega(n) \geq 2, \\ \dfrac{1}{2p-1}, & n = p^a,\ a \geq 2, \\ 0, & n = p, \end{cases} \tag{4.2}$$

and the optimal step is

$$\varepsilon^* = \begin{cases} \dfrac{2(p-1)(q-1)}{\varphi(n)\big(2(p-1)(q-1)+q-2\big)}, & \omega(n) \geq 2, \\ \dfrac{2(p-1)}{(2p-1)\,\varphi(n)}, & n = p^a,\ a \geq 2, \\ \dfrac{1}{n}, & n = p. \end{cases} \tag{4.3}$$

Proof. Write $\varphi = \varphi(n)$, $u = p-1$ and $v = q-1$. Suppose first $\omega(n) \geq 2$. By Theorem 3.2, $\lambda_{\max} = \varphi(u+1)/u$ and $\lambda_2 = \varphi(uv-1)/(uv)$. Therefore

$$\lambda_{\max} - \lambda_2 = \varphi\,\frac{(u+1)v-(uv-1)}{uv} = \varphi\,\frac{v+1}{uv},$$
$$\lambda_{\max} + \lambda_2 = \varphi\,\frac{(u+1)v+uv-1}{uv} = \varphi\,\frac{2uv+v-1}{uv},$$

and the quotient is $(v+1)/(2uv+v-1) = q/(2(p-1)(q-1)+q-2)$. Moreover $\varepsilon^* = 2/(\lambda_{\max}+\lambda_2) = 2uv/(\varphi(2uv+v-1))$. If $n = p^a$ with $a \geq 2$, then $\lambda_{\max} = \varphi(u+1)/u$ and $\lambda_2 = \varphi$, so $\lambda_{\max}-\lambda_2 = \varphi/u$, $\lambda_{\max}+\lambda_2 = \varphi(2u+1)/u$, and the quotient is $1/(2u+1) = 1/(2p-1)$, with $\varepsilon^* = 2u/(\varphi(2u+1))$. If $n = p$, then $\lambda_2 = \lambda_{\max} = n$, so $\rho^* = 0$ and $\varepsilon^* = 1/n$. The statement now follows from Lemma 4.1. □

Remark 4.4. The rate depends on n only through (p,q); the size of n, the multiplicities of its prime factors and all larger primes are irrelevant. For instance $\rho^*(X_6) = \rho^*(X_{30}) = \rho^*(X_{6^{100}}) = 3/5$. Table 1 lists some values.

Corollary 4.5 (Universal bound). *For every $n \geq 2$,*

$$\rho^*(X_n) \leq \frac{3}{5},$$

with equality if and only if $6 \mid n$. Consequently, for every n and every $\delta \in (0,1)$, the optimally tuned iteration (4.1) on X_n satisfies $\|e(t)\| \leq \delta\|e(0)\|$ as soon as

$$t \;\geq\; \frac{\log(1/\delta)}{\log(5/3)},$$

a bound that does not depend on n. For example, 28 iterations reduce the error by a factor 10^6 on every unitary Cayley graph.

Proof. Let $\omega(n) \geq 2$, so $p \geq 2$ and $q \geq 3$. A direct expansion gives

$$3\big(2(p-1)(q-1)+q-2\big) - 5q = 6(p-2)(q-3) + 12(p-2) + 4(q-3) \;\geq 0,$$

with equality if and only if $p = 2$ and $q = 3$. Since the denominator in (4.2) is positive, this is the claim for $\omega(n) \geq 2$, and $(p,q) = (2,3)$ means $6 \mid n$. For $n = p^a$ with $a \geq 2$ we have $\rho^* = 1/(2p-1) \leq 1/3 < 3/5$, and for n prime $\rho^* = 0$; in neither case does 6 divide n. The iteration count follows from $\|e(t)\| \leq (3/5)^t\|e(0)\|$, and $\log(10^6)/\log(5/3) \approx 27.05$. □

Remark 4.6 (Set of values). For fixed p the rate (4.2) decreases in q and tends to $1/(2p-1)$ as $q \to \infty$, which is the value at prime powers of p. Since moreover $1/(2p-1) \to 0$ as $p \to \infty$, the set $\{\rho^*(X_n) : n \geq 2\}$ is a countable subset of $[0, 3/5]$ that accumulates at 0 and at each of the numbers $1/(2p-1)$, p prime. Among even $n > 2$ the rate lies in $[1/3, 3/5]$, and among odd n it lies in $[0, 5/19]$, the value $5/19$ being attained exactly when $15 \mid n$.

The optimal step requires knowledge of p and q. The most natural untuned choice is $\varepsilon = 1/\varphi(n)$, for which $I - \varepsilon L$ is the transition matrix of the simple random walk on X_n.

Proposition 4.7 (Natural step). *With $\varepsilon = 1/\varphi(n)$ the iteration (4.1) on X_n has contraction factor*

$$\rho\big(1/\varphi(n)\big) = \frac{1}{p-1}.$$

It converges if and only if n is odd; for even n the graph is bipartite and the error does not decay in general.

Proof. The eigenvalues of $I - L/\varphi(n) = A/\varphi(n)$ on $\mathbf{1}^\perp$ are $r(M)$, $M \mid n$, $M > 1$. For such M let ℓ be a prime divisor of M; then $\varphi(\ell)$ divides $\varphi(M)$, so $\varphi(M) \geq \ell - 1 \geq p - 1$ and $|r(M)| \leq 1/(p-1)$, with equality at $M = p$. If $p = 2$ the eigenvalue $r(2) = -1$ belongs to the vector $\chi_{n/2} = ((-1)^j)_j$, whose component in the initial error is not damped. □

Remark 4.8 (Continuous time). For the flow $\dot{x} = -Lx$ on X_n one has $\|x(t) - \bar{x}\mathbf{1}\| \leq \mathrm{e}^{-\lambda_2 t}\|x(0) - \bar{x}\mathbf{1}\|$ with λ_2 given by (3.2). After normalising the coupling by the degree, the rate $\lambda_2/\varphi(n) = 1 - 1/((p-1)(q-1))$ is at least $1/2$ for all n, with equality if and only if $6 \mid n$.

Since $\rho^*(X_n)$ depends only on (p,q), its statistical behaviour over the integers can be described exactly.

Proposition 4.9 (Mean value). *The limit*

$$\bar{\rho} = \lim_{N\to\infty} \frac{1}{N} \sum_{n=2}^{N} \rho^*(X_n)$$

exists and is given by the convergent series

$$\bar{\rho} = \sum_{p<q} \frac{1}{pq} \prod_{\substack{\ell<q \\ \ell\neq p}} \Big(1 - \frac{1}{\ell}\Big) \cdot \frac{q}{2(p-1)(q-1)+q-2}, \tag{4.4}$$

where p, q and ℓ run over primes. Its numerical value is discussed in Remark 4.10.

Proof. For primes $p < q$ let $A_{p,q}$ be the set of n whose two smallest distinct prime divisors are p and q. Membership in $A_{p,q}$ is the conjunction of the conditions $p \mid n$, $q \mid n$ and $\ell \nmid n$ for all primes $\ell < q$, $\ell \neq p$. By the Chinese remainder theorem $A_{p,q}$ is a union of residue classes modulo $\prod_{\ell\le q}\ell$ and has natural density

$$d_{p,q} = \frac{1}{pq} \prod_{\substack{\ell<q \\ \ell\neq p}} \Big(1 - \frac{1}{\ell}\Big).$$

Fix $y \ge 3$ and let E_y be the set of n with at most one distinct prime divisor not exceeding y. The sets $A_{p,q}$ with $q \le y$ partition the complement of E_y. The set E_y is also a union of residue classes: either no prime $\ell \le y$ divides n, or exactly one prime $p \le y$ does, so that

$$d(E_y) = \prod_{\ell\le y}\Big(1-\frac{1}{\ell}\Big)\Big(1+\sum_{p\le y}\frac{1}{p-1}\Big).$$

By Mertens' theorems [8, §22.7–22.8] the product is $O(1/\log y)$ and the sum is $O(\log\log y)$, so $d(E_y) \to 0$ as $y \to \infty$. Since $\sum_{p<q\le y} d_{p,q} = 1 - d(E_y)$, we get $\sum_{p<q} d_{p,q} = 1$, and the series (4.4), whose terms are bounded by $\frac{3}{5}d_{p,q}$, converges.

Let $f(n) = \rho^*(X_n)$, so $0 \le f \le 3/5$ by Corollary 4.5 and f equals the constant $f_{p,q} = q/(2(p-1)(q-1)+q-2)$ on $A_{p,q}$. For every y,

$$\Big|\frac{1}{N}\sum_{n\le N} f(n) - \sum_{p<q\le y} f_{p,q}\,\frac{\#\{n\le N : n\in A_{p,q}\}}{N}\Big| \le \frac{3}{5}\cdot\frac{\#\{n\le N : n\in E_y\}}{N}.$$

Letting $N \to \infty$ with y fixed, the finitely many frequencies converge to $d_{p,q}$ and the right-hand side converges to $\frac{3}{5}d(E_y)$. Hence the upper and lower limits of $\frac{1}{N}\sum_{n\le N} f(n)$ differ from $\sum_{p<q\le y} f_{p,q}d_{p,q}$ by at most $\frac{3}{5}d(E_y)$. Letting $y \to \infty$ proves the existence of the limit and formula (4.4). The numerical value is discussed in Remark 4.10. □

Remark 4.10 (Numerical evaluation). The series (4.4) converges slowly, at the rate of $d(E_y)$. Writing $\delta_p = \frac{1}{p}\prod_{\ell<p}(1-\frac{1}{\ell})$ for the density of the integers with least prime divisor p and $w_{p,q} = \frac{1}{q}\prod_{p<\ell<q}(1-\frac{1}{\ell})$, so that $d_{p,q} = \delta_p w_{p,q}$ and $\sum_{q>p} w_{p,q} = 1$, and using

$$\frac{q}{2(p-1)(q-1)+q-2} - \frac{1}{2p-1} = \frac{2p}{(2p-1)\big(2(p-1)(q-1)+q-2\big)},$$

one obtains the rearrangement

$$\bar{\rho} = \sum_{p} \frac{\delta_p}{2p-1}\Big(1 + 2p\sum_{q>p}\frac{w_{p,q}}{2(p-1)(q-1)+q-2}\Big),$$

in which both sums converge rapidly. Truncating at primes below $2\cdot 10^6$ gives $\bar{\rho} \approx 0.27580$, and the empirical average of $\rho^*(X_n)$ over $2 \le n \le 5\cdot 10^6$ is 0.27580 as well.

5. The Kuramoto model on unitary Cayley graphs

Consider n identical Kuramoto oscillators coupled along a circulant graph $\mathrm{Cay}(\mathbb{Z}_n, S)$:

$$\dot\theta_i = \omega + \frac{K}{|S|}\sum_{s\in S}\sin\bigl(\theta_{i+s}-\theta_i\bigr), \qquad i\in\mathbb{Z}_n, \tag{5.1}$$

with common frequency $\omega\in\mathbb{R}$ and coupling strength $K>0$. Passing to the rotating frame $\theta_i\mapsto\theta_i-\omega t$ we may and do assume $\omega=0$. Then (5.1) is the gradient flow $\dot\theta=-\nabla E(\theta)$ of the energy

$$E(\theta) = -\frac{K}{2|S|}\sum_{i\in\mathbb{Z}_n}\sum_{s\in S}\cos\bigl(\theta_{i+s}-\theta_i\bigr), \tag{5.2}$$

in which every edge is counted twice. Since $S=-S$, the sum $\sum_i\theta_i$ is conserved, and E is invariant under $\theta\mapsto\theta+c\mathbf{1}$. Stability is always understood modulo this rotation, that is, for the flow restricted to an invariant hyperplane $\{\sum_i\theta_i=\mathrm{const}\}$.

Definition 5.1. For $m\in\mathbb{Z}_n$ the *m-twisted state* is $\theta^{(m)}_j = 2\pi mj/n$, $j\in\mathbb{Z}_n$. Its *order* is $M=\mathrm{ord}(m)$. The state $m=0$ is the synchronous state.

The following lemma is the specialisation to our normalisation of the linearisation carried out by Medvedev and Tang in [16, §3.1]; for nearest-neighbour rings see [23].

Lemma 5.2 (Twisted states and their Jacobians)**.** *Let $S=-S\subseteq\mathbb{Z}_n\setminus\{0\}$ and $m\in\mathbb{Z}_n$.*

(i) *$\theta^{(m)}$ is an equilibrium of (5.1) with $\omega=0$.*

(ii) *The Jacobian $J^{(m)}$ of (5.1) at $\theta^{(m)}$ is the symmetric circulant matrix with entries $J^{(m)}_{i,i+s}=\frac{K}{|S|}\cos\frac{2\pi ms}{n}$ for $s\in S$, $J^{(m)}_{ii}=-\frac{K}{|S|}C_S(m)$, and zero otherwise. Equivalently, $-J^{(m)}$ is the Laplacian of the graph $\mathrm{Cay}(\mathbb{Z}_n,S)$ with the edge weights $\frac{K}{|S|}\cos\frac{2\pi ms}{n}$.*

(iii) *$J^{(m)}\chi_k=\frac{K}{|S|}\,\mu_k(m)\,\chi_k$ for all $k\in\mathbb{Z}_n$, where*

$$\mu_k(m)=\tfrac12\bigl(C_S(m+k)+C_S(m-k)\bigr)-C_S(m). \tag{5.3}$$

In particular $\mu_0(m)=0$ and $\mu_{-k}(m)=\mu_k(m)$.

Proof. (i) At $\theta^{(m)}$ the right-hand side of (5.1) is $\frac{K}{|S|}\sum_{s\in S}\sin(2\pi ms/n)$ for every i, and the summand is odd under $s\mapsto -s$, which permutes S. (ii) Differentiating, $\partial\dot\theta_i/\partial\theta_{i+s}=\frac{K}{|S|}\cos(\theta_{i+s}-\theta_i)$ for $s\in S$ and $\partial\dot\theta_i/\partial\theta_i=-\frac{K}{|S|}\sum_{s\in S}\cos(\theta_{i+s}-\theta_i)$; at $\theta^{(m)}$ the phase difference is $2\pi ms/n$. (iii) Applying $J^{(m)}$ to χ_k gives the eigenvalue $\frac{K}{|S|}\sum_{s\in S}\cos\frac{2\pi ms}{n}\bigl(\mathrm{e}^{2\pi iks/n}-1\bigr)$. The imaginary part vanishes by the symmetry $s\mapsto -s$, and for the real part we use

$$\cos a\,(\cos b-1)=\tfrac12\bigl(\cos(a+b)+\cos(a-b)\bigr)-\cos a \tag{5.4}$$

with $a=2\pi ms/n$, $b=2\pi ks/n$, and sum over $s\in S$ using (2.2). □

For X_n we have $|S|=\varphi(n)$ and $C_S(j)=c_n(j)=\varphi(n)r(\mathrm{ord}(j))$ by (2.3), so that

$$\frac{\mu_k(m)}{\varphi(n)}=\tfrac12\bigl(r(\mathrm{ord}(m+k))+r(\mathrm{ord}(m-k))\bigr)-r(\mathrm{ord}(m)). \tag{5.5}$$

5.1. **The instability exponent.** The choice $k = m$ in (5.5) produces a quantity that depends only on the order of the twist. If $\operatorname{ord}(m) = M$, then $\operatorname{ord}(2m) = M/\gcd(2, M)$, and $\operatorname{ord}(0) = 1$. We define, for integers $M \geq 1$,

$$g(M) = \tfrac{1}{2}\Big(1 + r\Big(\frac{M}{\gcd(2, M)}\Big)\Big) - r(M), \tag{5.6}$$

so that $\mu_m(m) = \varphi(n)\, g(\operatorname{ord}(m))$ for every $m \in \mathbb{Z}_n$.

Lemma 5.3 (Sign of the exponent)**.** *We have $g(1) = 0$, $g(4) = 0$, $g(6) = -\tfrac{1}{4}$, and $g(M) \geq \tfrac{1}{8}$ for every $M \notin \{1, 4, 6\}$. Moreover $g(2) = 2$.*

Proof. Clearly $g(1) = \tfrac{1}{2}(1 + 1) - 1 = 0$. Let $M > 1$.

Case 1: M odd. Then $g(M) = \tfrac{1}{2}(1 + r(M)) - r(M) = \tfrac{1}{2}(1 - r(M))$. Since $M \geq 3$ we have $|r(M)| \leq \tfrac{1}{2}$, so $g(M) \geq \tfrac{1}{4}$.

Case 2: $M = 2M'$ with M' odd. Then $\mu(M) = -\mu(M')$ and $\varphi(M) = \varphi(M')$, so $r(M) = -r(M')$ and

$$g(M) = \tfrac{1}{2}\big(1 + r(M')\big) + r(M') = \tfrac{1}{2} + \tfrac{3}{2}\, r(M').$$

If $M' = 1$, then $g(2) = 2$. If $M' = 3$, then $r(3) = -\tfrac{1}{2}$ and $g(6) = -\tfrac{1}{4}$. Let $M' \geq 5$ be odd. If $r(M') \geq 0$ then $g(M) \geq \tfrac{1}{2}$. If $r(M') < 0$, then M' is squarefree, and since $M' \neq 3$ it has a prime divisor $\ell \geq 5$; as $\varphi(\ell)$ divides $\varphi(M')$ we get $\varphi(M') \geq 4$, so $r(M') \geq -\tfrac{1}{4}$ and $g(M) \geq \tfrac{1}{2} - \tfrac{3}{8} = \tfrac{1}{8}$.

Case 3: $4 \mid M$. Then $\mu(M) = 0$ and $g(M) = \tfrac{1}{2}(1 + r(M/2))$ with $M/2$ even. If $M = 4$, then $r(2) = -1$ and $g(4) = 0$. If $M \geq 8$, then $M/2 \geq 4$, so $|r(M/2)| \leq \tfrac{1}{2}$ and $g(M) \geq \tfrac{1}{4}$. □

The values $g(M)$ for $2 \leq M \leq 60$ are displayed in Figure 4.

5.2. **Classification.** We now present the main results of this section.

Theorem 5.4 (Classification of twisted states on X_n)**.** *Let $n \geq 2$, let $m \in \mathbb{Z}_n \setminus \{0\}$, and let $M = \operatorname{ord}(m)$. Consider the m-twisted state of (5.1) on X_n.*

(a) *If $M = 6$, then $\mu_k(m) \leq -\varphi(n)/4$ for all $k \neq 0$; the state is linearly stable.*
(b) *If $M = 4$, then $J^{(m)} = 0$.*
(c) *If $M \notin \{4, 6\}$, then $J^{(m)}$ has the positive eigenvalue $K\, g(M) \geq K/8$ with eigenvector $\chi_m \perp \mathbf{1}$; the state is linearly unstable.*

Proof. (c) Since $m \neq 0$, the index $k = m$ is non-zero and $\chi_m \perp \mathbf{1}$. By Lemma 5.2(iii) and the definition of g, the corresponding eigenvalue is $\frac{K}{\varphi(n)}\mu_m(m) = K\, g(M)$, and $g(M) \geq \tfrac{1}{8}$ by Lemma 5.3 because $M \notin \{1, 4, 6\}$. The real and imaginary parts of χ_m are real eigenvectors orthogonal to $\mathbf{1}$, at least one of which is non-zero, so the equilibrium is unstable by Lyapunov's first method [12, Thm. 4.7].

Parts (a) and (b) are contained in Theorems 5.5 and 5.6 below, where more is proved. □

Theorem 5.5 (Hexagonal states)**.** *Let $6 \mid n$ and $m \in \{n/6, 5n/6\}$, so that $\theta_j^{(m)} = \pm j\pi/3$. Then*

$$J^{(m)} = -\frac{K}{2\varphi(n)}\, L, \tag{5.7}$$

where L is the Laplacian of X_n. The eigenvalues of $J^{(m)}$ on $\mathbf{1}^{\perp}$ are $-\frac{K}{2}\big(1 - r(M')\big)$, $M' \mid n$, $M' > 1$; they lie in $[-K, -K/4]$, the value $-K/4$ being attained exactly on the two-dimensional space spanned by $\chi_{\pm n/6}$. Consequently $\theta^{(m)}$ is exponentially stable modulo rotation, with linear decay rate exactly $K/4$, which is one half of the rate $K/2$ of the synchronous state on the same graph.

Proof. Let s be a unit modulo n. Since $6 \mid n$, s is coprime to 6, so $s \equiv \pm 1 \pmod 6$ and

$$\cos\frac{2\pi m s}{n} = \cos\Big(\pm\frac{\pi s}{3}\Big) = \cos\frac{\pi}{3} = \frac{1}{2}.$$

By Lemma 5.2(ii), $-J^{(m)}$ is the Laplacian of X_n with all edge weights equal to $K/(2\varphi(n))$, which is (5.7). The spectrum follows from (2.4). Since $p = 2$ and $q = 3$, Theorem 3.2 gives $\lambda_2 = \varphi(n)/2$ with multiplicity $(p-1)(q-1) = 2$, attained at the indices of order 6, namely $\pm n/6$, and $\lambda_{\max} = 2\varphi(n)$; multiplying by $-K/(2\varphi(n))$ gives the interval $[-K, -K/4]$. At the synchronous state all weights equal $K/\varphi(n)$, so $J^{(0)} = -\frac{K}{\varphi(n)}L$ and the slowest rate is $K\lambda_2/\varphi(n) = K/2$. Finally, on the invariant hyperplane through $\theta^{(m)}$ the equilibrium is hyperbolic with all eigenvalues at most $-K/4$, which gives local exponential stability with every rate below $K/4$ by Lyapunov's indirect method [12, Thm. 4.7]. □

Theorem 5.6 (Quarter twists)**.** *Let $4 \mid n$ and $m \in \{n/4,\, 3n/4\}$, so that $\theta^{(m)}_j = \pm j\pi/2$. Then:*

(i) *$J^{(m)} = 0$ and $E(\theta^{(m)}) = 0$.*

(ii) *For $t \in \mathbb{R}$ let $\delta^{(t)} = t(e_0 - e_1)$, where e_j is the j-th standard basis vector. Then*

$$E\big(\theta^{(m)} + \delta^{(t)}\big) = \sigma\,\frac{2K}{\varphi(n)}\,\sin t\,(1-\cos t), \qquad \sigma = \sin\frac{2\pi m}{n} \in \{1,-1\}. \tag{5.8}$$

(iii) *$\theta^{(m)}$ is not a local minimum of E, even along perturbations of zero sum. Consequently, by the theorem of Absil and Kurdyka [1] on analytic gradient flows, $\theta^{(m)}$ is unstable; in particular it is not asymptotically stable.*

Proof. Write $m = an/4$ with $a \in \{1, 3\}$ and put $\sigma(s) = \sin(\pi a s/2)$ for $s \in \mathbb{Z}$. As n is even, every unit s is odd, so $\cos(2\pi m s/n) = \cos(\pi a s/2) = 0$ and $\sigma(s) \in \{1, -1\}$; moreover $\sigma(-s) = -\sigma(s)$, so $\sum_{s\in\mathbb{Z}_n^\times} \sigma(s) = 0$. Note that $\sigma(s)$ depends only on s modulo 4 and $4 \mid n$, so σ is well defined on $\mathbb{Z}_n^\times$.

(i) All edge weights in Lemma 5.2(ii) vanish, hence $J^{(m)} = 0$; and every term of (5.2) is $\cos(\pi a s/2) = 0$.

(ii) For a perturbation δ, using $\cos(\alpha+\beta) = -\sin\alpha\sin\beta$ when $\cos\alpha = 0$,

$$E\big(\theta^{(m)} + \delta\big) = \frac{K}{2\varphi(n)} \sum_{i\in\mathbb{Z}_n} \sum_{s\in\mathbb{Z}_n^\times} \sigma(s)\,\sin(\delta_{i+s} - \delta_i).$$

Let $U = \mathbb{Z}_n^\times$ and $\delta = \delta^{(t)}$, so $\delta_0 = t$, $\delta_1 = -t$ and $\delta_j = 0$ otherwise. Note $1 \in U$ and $0 \notin U$. We evaluate the double sum Σ according to i.

For $i = 0$: the term $s = 1$ contributes $\sigma(1)\sin(-2t)$, and each $s \in U \setminus \{1\}$ contributes $\sigma(s)\sin(-t)$. Since $\sum_{s\in U\setminus\{1\}} \sigma(s) = -\sigma(1)$, the total is $-\sigma(1)\sin 2t + \sigma(1)\sin t$.

For $i = 1$: the term $s = -1$ contributes $\sigma(-1)\sin(2t) = -\sigma(1)\sin 2t$, and each $s \in U \setminus \{-1\}$ contributes $\sigma(s)\sin t$. Since $\sum_{s\in U\setminus\{-1\}} \sigma(s) = -\sigma(-1) = \sigma(1)$, the total is $-\sigma(1)\sin 2t + \sigma(1)\sin t$.

For $i \notin \{0, 1\}$: only $s = -i$ (if $i \in U$) and $s = 1-i$ (if $1-i \in U$) give non-zero terms, namely $\sigma(-i)\sin t = -\sigma(i)\sin t$ and $\sigma(1-i)\sin(-t)$. Summing over $i \notin \{0,1\}$, the first kind gives $-\sin t \sum_{i\in U\setminus\{1\}} \sigma(i) = \sigma(1)\sin t$, and the second kind, with $j = 1-i$ running over $U \setminus \{1\}$, gives $-\sin t \sum_{j\in U\setminus\{1\}} \sigma(j) = \sigma(1)\sin t$.

Altogether $\Sigma = \sigma(1)\big(4\sin t - 2\sin 2t\big) = 4\sigma(1)\sin t\,(1-\cos t)$, and $\sigma(1) = \sin(\pi a/2) = \sin(2\pi m/n)$. Multiplying by $K/(2\varphi(n))$ gives (5.8).

(iii) The vector $\delta^{(t)}$ has zero sum, and by (5.8) the energy difference $E(\theta^{(m)} + \delta^{(t)}) - E(\theta^{(m)})$ has the sign of σt for $0 < |t| < \pi$. Hence every neighbourhood of $\theta^{(m)}$, even within its invariant

hyperplane, contains points of strictly smaller energy, and $\theta^{(m)}$ is not a local minimum of E. The energy E is real analytic, and (5.1) is the gradient flow $\dot\theta = -\nabla E$ of a C^1, globally Lipschitz vector field. For such flows Lyapunov stability of an equilibrium implies that it is a local minimum [1]; hence $\theta^{(m)}$ is unstable, and a fortiori not asymptotically stable. □

Corollary 5.7 (The role of 6)**.** *The twisted states of the Kuramoto model on X_n that are linearly stable are the synchronous state and, when $6 \mid n$, the two hexagonal states $m = \pm n/6$. In particular X_n admits a stable non-synchronous twisted state if and only if $6 \mid n$.*

Proof. The synchronous state has $J^{(0)} = -\frac{K}{\varphi(n)}L$, which is negative definite on $\mathbf{1}^\perp$ because X_n is connected. For $m \neq 0$ apply Theorems 5.4, 5.5 and 5.6: the state is stable for $M = 6$ and unstable for all other M. Elements of order 6 exist in $\mathbb{Z}_n$ if and only if $6 \mid n$, and then they are $\pm n/6$. □

Remark 5.8 (Phase-cohesive interpretation). At a hexagonal state every edge of X_n carries the phase difference $\pm\pi/3$: the vertices fall into six clusters according to $j \bmod 6$, and units are $\equiv \pm 1 \pmod 6$. This is why the standard sufficient condition for stability, that all phase differences across edges are smaller than $\pi/2$ in absolute value [5], applies. Theorem 5.4 shows that on X_n this sufficient condition is also necessary among twisted states, apart from the degenerate order $M = 4$ in which every edge carries exactly $\pm\pi/2$.

Remark 5.9 (Density). The graph X_n has relative degree $\varphi(n)/(n-1)$, which for $6 \mid n$ is at most $n/(3(n-1))$, with equality exactly when 2 and 3 are the only prime divisors of n; in particular it is below $3/4$. This is consistent with the theorem of Kassabov, Strogatz and Townsend [10] that networks in which every oscillator is coupled to at least 75% of the others synchronise globally, and it adds an arithmetic family to the dense circulant examples with stable non-synchronous states constructed in [22].

6. GCD graphs

Much of the preceding analysis rests on statements valid for arbitrary circulant graphs, which we isolate here and then apply to gcd graphs.

Proposition 6.1 (General circulant criteria)**.** *Let $G = \mathrm{Cay}(\mathbb{Z}_n, S)$ with $S = -S$, and let $m \in \mathbb{Z}_n \setminus \{0\}$. In part (ii) assume that G is connected, that is, that S generates $\mathbb{Z}_n$.*

(i) *If $S \neq \emptyset$ and $C_S(m) < \frac{1}{2}\big(|S| + C_S(2m)\big)$, then the m-twisted state is linearly unstable, and hence unstable. In particular this holds whenever $C_S(m) < 0$.*

(ii) *If $\cos(2\pi ms/n) > 0$ for every $s \in S$, then the m-twisted state is linearly stable.*

Proof. (i) By (5.3) with $k = m$, $\mu_m(m) = \frac{1}{2}(|S| + C_S(2m)) - C_S(m) > 0$, and $\chi_m \perp \mathbf{1}$. For the last assertion, $C_S(2m) \geq -|S|$ by (2.2), so the right-hand side of the hypothesis is non-negative. Instability of the nonlinear system follows as in Theorem 5.4. (ii) By Lemma 5.2(ii), $-J^{(m)}$ is the Laplacian of G with strictly positive edge weights c_s, so $\langle -J^{(m)}w, w\rangle = \frac{K}{2|S|}\sum_i \sum_{s\in S} c_s(w_{i+s} - w_i)^2 \geq 0$. If this vanishes, then $w_{i+s} = w_i$ for all i and $s \in S$, hence for all s in the subgroup generated by S, which is $\mathbb{Z}_n$; so w is constant, and $w = 0$ if $w \perp \mathbf{1}$. Thus $J^{(m)}$ is negative definite on $\mathbf{1}^\perp$. □

Part (i) says that a twisted state whose index m labels a negative adjacency eigenvalue is always unstable. For X_n this already disposes of all m with $\mu(\mathrm{ord}(m)) = -1$.

Proposition 6.2 (Hexagonal states on gcd graphs)**.** *Let $6 \mid n$ and let D be a set of proper divisors of n, each coprime to 6, such that $X_n(D)$ is connected. Then at $m = \pm n/6$ the Jacobian of (5.1) on $X_n(D)$ is*

$$J^{(m)} = -\frac{K}{2|S_D|}\, L_D,$$

where L_D is the Laplacian of $X_n(D)$. Consequently $\lambda_2(L_D) > 0$, the linearisation at the hexagonal states decays at the rate $K\lambda_2(L_D)/(2|S_D|)$, one half of the corresponding rate $K\lambda_2(L_D)/|S_D|$ of the synchronous state, and both states are locally exponentially stable modulo rotation with every rate below these values.

Proof. If $\gcd(s,n) = d \in D$ and d is coprime to 6, then, because $6 \mid n$, neither 2 nor 3 divides s; hence $s \equiv \pm 1 \pmod 6$ and $\cos(2\pi m s/n) = \cos(\pi s/3) = \frac{1}{2}$. The formula for $J^{(m)}$ follows from Lemma 5.2(ii) as in the proof of Theorem 5.5. For $k \neq 0$ we have $C_{S_D}(k) < |S_D|$, since equality would force $\mathrm{e}^{2\pi i k s/n} = 1$ for all $s \in S_D$, hence for all s in the subgroup they generate, which is $\mathbb{Z}_n$, so $k = 0$. Thus $\lambda_2(L_D) > 0$, and the rates follow from Lyapunov's indirect method [12, Thm. 4.7] as in Theorem 5.5. □

Remark 6.3. The same argument shows that if $4 \mid n$ and all elements of D are odd, then the Jacobian at the quarter twists of $X_n(D)$ vanishes. More generally, at the hexagonal state an edge of step s carries the weight $\cos(\pi s/3)$, which equals $\frac{1}{2}$, $-\frac{1}{2}$, -1 or 1 according as $\gcd(s,6)$ is 1, 2, 3 or 6. If D contains a divisor d with $\gcd(d,6) \in \{2,3\}$, edges with the negative weights $-\frac{1}{2}$ or -1 appear, and stability must then be decided from (5.3) and (2.5).

Proposition 6.4 (Consensus on gcd graphs)**.** *Let $n \geq 2$. Then $\rho^*(X_n(D))$ is a rational number, given by*

$$\rho^*\big(X_n(D)\big) = \frac{C^+ - C^-}{2|S_D| - C^+ - C^-}, \qquad C^+ = \max_{k\neq 0} \sum_{d\in D} c_{n/d}(k), \quad C^- = \min_{k\neq 0} \sum_{d\in D} c_{n/d}(k).$$

Proof. By (2.5) the Laplacian eigenvalues are $|S_D| - C_{S_D}(k)$. They are integers, because each Ramanujan sum $c_N(j) = \varphi(N)\mu(M)/\varphi(M)$ is an integer, $\varphi(M)$ dividing $\varphi(N)$ for $M \mid N$; thus $\lambda_2 = |S_D| - C^+$ and $\lambda_{\max} = |S_D| - C^-$, and the formula follows from Lemma 4.1. □

7. Numerical illustrations

All statements above were checked by direct computation; the scripts are part of the supplementary material. For every $2 \leq n \leq 260$ the Laplacian of X_n was diagonalised numerically and λ_2, $\lambda_{\max}$ and ρ^* were compared with (3.2), (3.1) and (4.2); the bound of Corollary 4.5 and its equality case were checked for all $n \leq 2 \cdot 10^5$. For every $2 \leq n \leq 150$ and every $m \neq 0$ the Jacobian $J^{(m)}$ was assembled and its spectrum compared with (5.5); among the 11175 pairs (n,m) there are 50 hexagonal states, 74 quarter twists and 11051 unstable states, in agreement with Theorem 5.4, and the formula (5.8) was confirmed for several n. No discrepancy was found.

Figure 1 shows $\rho^*(X_n)$ for $n \leq 300$ and makes the structure described in Remark 4.6 visible. Figure 2 shows the error of the optimally tuned iteration (4.1) for $n = 30, 32, 35$ from random initial data together with the predicted envelopes $(\rho^*)^t$; for $n = 30$ the error stays below the envelope without being parallel to it at first, because a random initial vector has only a small component along the slowest modes. Figure 3 illustrates Theorems 5.4 and 5.5 on X_{30} with $K = 1$: a perturbation of size 10^{-2} of the hexagonal state $m = 5$ decays at the rate $K/4$, the same perturbation of the synchronous state decays at the rate $K/2$, and perturbed twisted states of orders 10 and 30 leave their neighbourhood and reach synchrony.

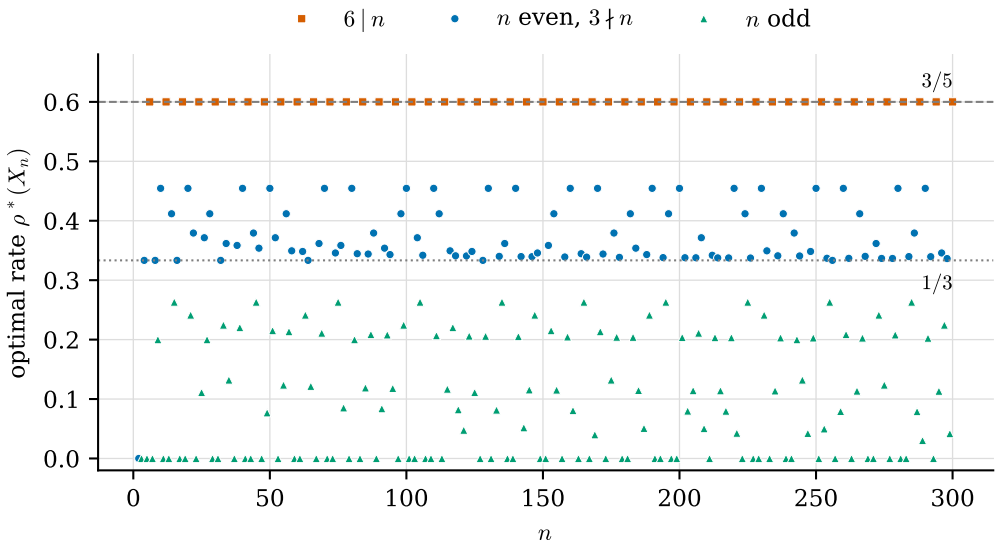


FIGURE 1. The optimal consensus rate $\rho^*(X_n)$ for $2 \le n \le 300$. The supremum $3/5$ is attained exactly on the multiples of 6; even $n > 2$ not divisible by 3 lie in $[1/3, 5/11]$ and odd n in $[0, 5/19]$.

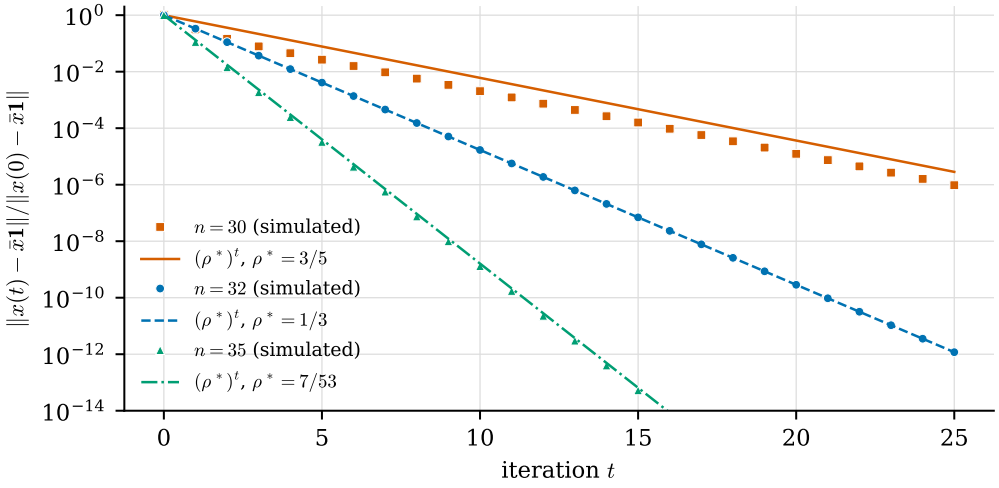


FIGURE 2. Relative consensus error for the optimal step (4.3) from random initial data (markers) and the bound $(\rho^*)^t$ of Theorem 4.3 (lines).

8. LEAN 4 FORMALISATION

The paper is accompanied by a Lean 4 project [17], `ArithmeticDynamics`, built on Mathlib [15], together with a single-file version of the same development. It consists of 19 parts with about 5400 lines and 312 theorems. It compiles with Lean `v4.36.0-rc2` against Mathlib (commit `2ddf59c8f142`), it contains no `sorry`, and every theorem depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`.

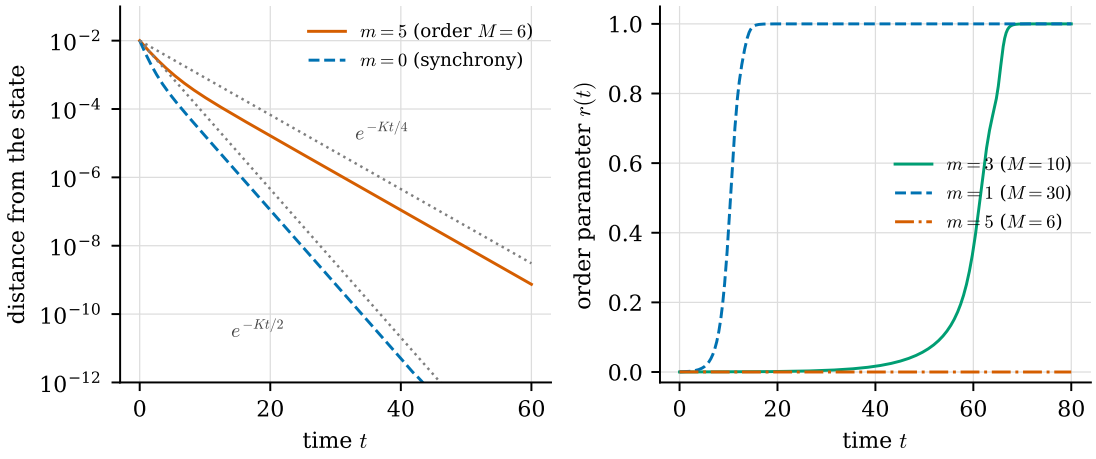


FIGURE 3. Kuramoto dynamics on X_{30} with $K = 1$. Left: distance to the hexagonal state $m = 5$ and to the synchronous state after a perturbation of norm 10^{-2}, with the predicted rates $K/4$ and $K/2$. Right: order parameter $r(t) = |n^{-1}\sum_j \mathrm{e}^{\mathrm{i}\theta_j}|$ for perturbed twisted states of orders 6, 10 and 30.

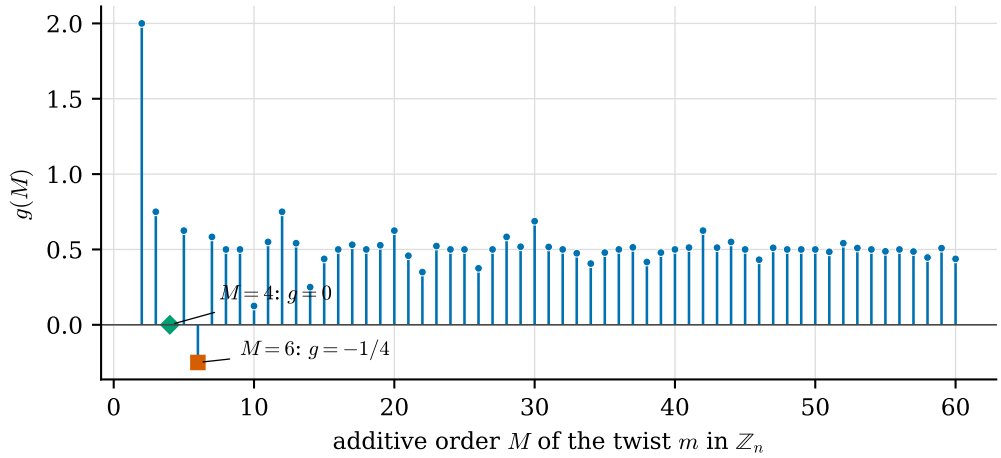


FIGURE 4. The exponent $g(M)$ of (5.6): the Jacobian at a twisted state of order M has the eigenvalue $K\,g(M)$. It is negative only for $M = 6$ and vanishes only for $M = 4$ (Lemma 5.3).

Scope. Except for three classical theorems from the stability theory of ordinary differential equations, described below, which enter as explicit hypotheses, every theorem, lemma, proposition and corollary of Sections 3–6, the identities (2.2)–(2.5) on which they rest, and every mathematical assertion made in the remarks has been formalised. Two kinds of material are outside the formalisation: the numerical computations of Section 7 and the decimal evaluation of $\bar{\rho}$ in Remark 4.10, which are computational; and background facts that are quoted in Section 2 but not used, such as So's characterisation of integral circulant graphs. Connectivity of a circulant graph is expressed in Lean by the condition that S generates $\mathbb{Z}_n$, which is equivalent

to it. Mean values are taken over $0 \le n < N$ instead of $2 \le n \le N$, which does not change the limit.

The classical hypotheses. The three classical theorems are stated as propositions about *every* vector field F on $\mathbb{R}^{\mathbb{Z}_n}$ that is C^1 and globally Lipschitz, the last condition guaranteeing that solutions exist for all time. Each is therefore a special case of a standard theorem, and none is tailored to the Kuramoto model. Stability is measured in the Euclidean norm, and a solution is a differentiable map $x : \mathbb{R} \to \mathbb{R}^{\mathbb{Z}_n}$ with $\dot{x} = F(x)$.

(H1) (Lean name `LyapunovIndirectStability`.) If F has zero coordinate sum, $F(x^*) = 0$, and $\langle DF(x^*)w, w\rangle \le -r\|w\|^2$ for all $w \perp \mathbf{1}$, then for every $0 < r' < r$ there is $\delta > 0$ such that every solution with $\sum_i x_i(0) = \sum_i x_i^*$ and $\|x(0) - x^*\|^2 < \delta$ satisfies $\|x(t) - x^*\|^2 \le \mathrm{e}^{-2r't}\|x(0) - x^*\|^2$ for $t \ge 0$. This is Lyapunov's indirect method [12, Thm. 4.7] on an invariant level set, with $\|x - x^*\|^2$ as Lyapunov function.

(H2) (Lean name `LyapunovIndirectInstability`.) If $F(x^*) = 0$ and $DF(x^*)v = \mu v$ with $v \ne 0$ and $\mu > 0$, then x^* is not Lyapunov stable [12, Thm. 4.7].

(H3) (Lean name `AbsilKurdyka`.) If E is real analytic and $F = -\nabla E$, then every Lyapunov-stable equilibrium of $\dot{x} = F(x)$ is a local minimum of E [1].

Hypothesis (H1) is used for the exponential stability statements of Theorem 5.5 and Proposition 6.2, (H2) for the nonlinear instability statements of Theorem 5.4(c) and Proposition 6.1(i), and (H3) for Theorem 5.6(iii). All the facts about the Kuramoto field that these hypotheses require are proved in Lean: that the field is C^1, globally Lipschitz and of zero sum, that twisted states are equilibria, the Fréchet derivative at twisted states, the analyticity of the energy, and the identity $F = -\nabla E$.

Design. Functions on the vertex set are maps $\mathbb{Z}_n \to \mathbb{C}$ or $\mathbb{Z}_n \to \mathbb{R}$, with `ZMod n` for $\mathbb{Z}_n$; X_n is the `SimpleGraph unitaryCayley n`, and general circulants are given by a symmetric `Finset` S. Characters come from Mathlib's `ZMod.stdAddChar`. Mathlib contains neither Ramanujan sums nor Parseval's identity for the discrete Fourier transform on $\mathbb{Z}_n$, so both are proved here: Hölder's formula by pushing the sum over units forward along $(\mathbb{Z}/n)^\times \to (\mathbb{Z}/M)^\times$ and applying Möbius inversion, and Plancherel's identity from the orthogonality of characters. Plancherel's identity converts every spectral statement into quadratic-form bounds on sum-zero vectors, which give the norm estimates of Section 4 and the stability inequalities of Section 5. The mean value of Proposition 4.9 is proved by computing densities through the Chinese remainder theorem (`ZMod.prodEquivPi`) and bounding the exceptional set by $\mathrm{e}^{-S}(1 + 2S)$ with $S = \sum_{p \le y} 1/p$, which tends to 0 by the divergence of $\sum 1/p$ in Mathlib. Mertens' theorems are therefore not needed.

9. Conclusion

We have shown that two standard dynamical processes on unitary Cayley graphs admit exact arithmetic descriptions. For linear consensus the optimal contraction factor is a rational function of the two smallest prime divisors of the modulus; it is bounded by $3/5$ uniformly in n, so that these graphs form a family on which consensus is reached in a number of steps independent of the size, and its average over all moduli is a computable constant close to 0.276. For the Kuramoto model the stability of every twisted state is decided by the additive order of the twist alone, and the only stable non-synchronous states are the hexagonal twists, present exactly when the modulus is divisible by six. Both statements are consequences of a single extremal property of the function $\mu(M)/\varphi(M)$ on the divisors of n, and the appearance of the

TABLE 2. Results of the paper and the principal Lean theorems proving them.

Result	Lean theorem(s)
(2.3), (2.4)	`ramanujan_eq`, `lap_chi`, `lap_eigenvalue`
(2.5)	`fiber_sum_eq_ramanujan`, `cSym_gcd_int`
Lemma 3.1	`isLeast_divSet`, `isGreatest_divSet_*`, `r_eq_min_iff`, `r_eq_max_iff`
Theorem 3.2	`lapEig_le`, `lapEig_ge_*`, `card_lapEig_max`, `card_lapEig_*`
Corollary 3.3	`spectral_ratio`, `spectral_ratio_formula`
Lemma 4.1	`contraction_of_opt_step`, `rate_lower_bound`, `sqNorm_err_iterate`
Theorem 4.3	`isRate_of_twoSmallest`, `isRate_prime_pow`, `isRate_prime`, `opt_step_*`, `consensus_of_isRate`
Corollary 4.5	`rate_le_and_eq_iff`, `consensus_three_fifths`, `three_fifths_pow_le`
Remark 4.6	`rate_strictAnti_q`, `rate_tendsto_q`, `rate_even`, `rate_odd`
Proposition 4.7	`natural_step_factor`, `natural_step_lt_one_iff`, `natural_step_even`
Remark (continuous time)	`continuous_consensus`
Proposition 4.9	`mean_value`, `density_inA`, `density_inE_le`
Lemma 5.2	`kuramotoField_twisted`, `fderiv_kuramotoField`, `jac_chi`
Lemma 5.3	`g_one`, `g_two`, `g_four`, `g_six`, `g_ge`
Theorem 5.4	`twisted_classification`, `twisted_unstable_nonlinear` (H2)
Theorem 5.5	`jac_hex`, `hex_spectrum`, `hex_expStable` (H1), `sync_expStable` (H1)
Theorem 5.6	`jac_quarter`, `energy_quarter_kick`, `quarter_not_isLocalMin`, `quarter_unstable` (H3)
Corollary 5.7	`linearly_stable_iff`, `exists_order_six_iff`
Remarks in Section 5	`twistWeight_hex`, `totient_le_third`, `relative_degree_lt`
Proposition 6.1	`circulant_unstable_linear`, `circulant_unstable` (H2), `circulant_stable_linear`
Proposition 6.2	`cJac_hex_gcd`, `gcd_hex_expStable` (H1), `gcd_sync_expStable` (H1)
Remark in Section 6	`cJac_quarter_gcd`, `cos_hex_cases`
Proposition 6.4	`gcd_rate`, `ramanujan_int`

number six in both is explained by the fact that six is the only integer greater than one at which this function reaches the value one half.

Several directions remain open. The quarter twists are equilibria at which the linearisation carries no information, and although we have shown that they are not minima of the energy, the local structure of the flow near them, and in particular the dimension of their stable sets, has not been determined. The basins of attraction of the hexagonal states are not known; numerical experiments suggest that on X_n with $6 \mid n$ a positive but small fraction of random initial conditions is attracted to them, and a quantitative statement would be of interest in connection with the density thresholds for global synchronisation. For general gcd graphs we have only treated divisor sets coprime to six, and a classification of the stable twisted states of $X_n(D)$ in terms of D seems to require new ideas when edges with negative weights are present. Finally, the Lean development proves every result of the paper modulo three classical theorems on the stability of equilibria of ordinary differential equations; formalising those theorems in Mathlib would remove the remaining hypotheses, and contributing the material on Ramanujan sums, circulant spectra and densities defined by congruence conditions to Mathlib is a natural continuation.

Data and code availability. The Lean 4 sources, the verification script and the scripts that generate all figures are supplied with the paper.

References

[1] P.-A. Absil and K. Kurdyka, On the stable equilibrium points of gradient systems, *Systems Control Lett.* **55** (2006), 573–577.

[2] M. Bašić, M. D. Petković and D. Stevanović, Perfect state transfer in integral circulant graphs, *Appl. Math. Lett.* **22** (2009), 1117–1121.

[3] S. Boyd, P. Diaconis and L. Xiao, Fastest mixing Markov chain on a graph, *SIAM Rev.* **46** (2004), 667–689.

[4] I. J. Dejter and R. E. Giudici, On unitary Cayley graphs, *J. Combin. Math. Combin. Comput.* **18** (1995), 121–124.

[5] F. Dörfler and F. Bullo, Synchronization in complex networks of phase oscillators: a survey, *Automatica* **50** (2014), 1539–1564.

[6] M. Fiedler, Algebraic connectivity of graphs, *Czechoslovak Math. J.* **23** (1973), 298–305.

[7] C. Godsil and G. Royle, *Algebraic Graph Theory*, Graduate Texts in Mathematics 207, Springer, New York, 2001.

[8] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, Oxford, 2008.

[9] A. Ilić, The energy of unitary Cayley graphs, *Linear Algebra Appl.* **431** (2009), 1881–1889.

[10] M. Kassabov, S. H. Strogatz and A. Townsend, Sufficiently dense Kuramoto networks are globally synchronizing, *Chaos* **31** (2021), 073135.

[11] W. Klotz and T. Sander, Some properties of unitary Cayley graphs, *Electron. J. Combin.* **14** (2007), #R45.

[12] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, Upper Saddle River, NJ, 2002.

[13] Y. Kuramoto, *Chemical Oscillations, Waves, and Turbulence*, Springer, Berlin, 1984.

[14] X. Liu and S. Zhou, Eigenvalues of Cayley graphs, *Electron. J. Combin.* **29**(2) (2022), #P2.9.

[15] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, ACM, 2020, pp. 367–381.

[16] G. S. Medvedev and X. Tang, Stability of twisted states in the Kuramoto model on Cayley and random graphs, *J. Nonlinear Sci.* **25** (2015), 1169–1208.

[17] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Comput. Sci. 12699, Springer, 2021, pp. 625–635.

[18] R. Olfati-Saber, J. A. Fax and R. M. Murray, Consensus and cooperation in networked multi-agent systems, *Proc. IEEE* **95** (2007), 215–233.

[19] S. Ramanujan, On certain trigonometrical sums and their applications in the theory of numbers, *Trans. Cambridge Philos. Soc.* **22** (1918), 259–276.

[20] N. Saxena, S. Severini and I. E. Shparlinski, Parameters of integral circulant graphs and periodic quantum dynamics, *Int. J. Quantum Inf.* **5** (2007), 417–430.

[21] W. So, Integral circulant graphs, *Discrete Math.* **306** (2006), 153–158.

[22] A. Townsend, M. Stillman and S. H. Strogatz, Dense networks that do not synchronize and sparse ones that do, *Chaos* **30** (2020), 083142.

[23] D. A. Wiley, S. H. Strogatz and M. Girvan, The size of the sync basin, *Chaos* **16** (2006), 015103.

[24] L. Xiao and S. Boyd, Fast linear iterations for distributed averaging, *Systems Control Lett.* **53** (2004), 65–78.

Department of Mathematics, University of Kashmir, South Campus, Anantnag, 192101, Jammu and Kashmir, India

Email address: sajadsheikh@uok.edu.in